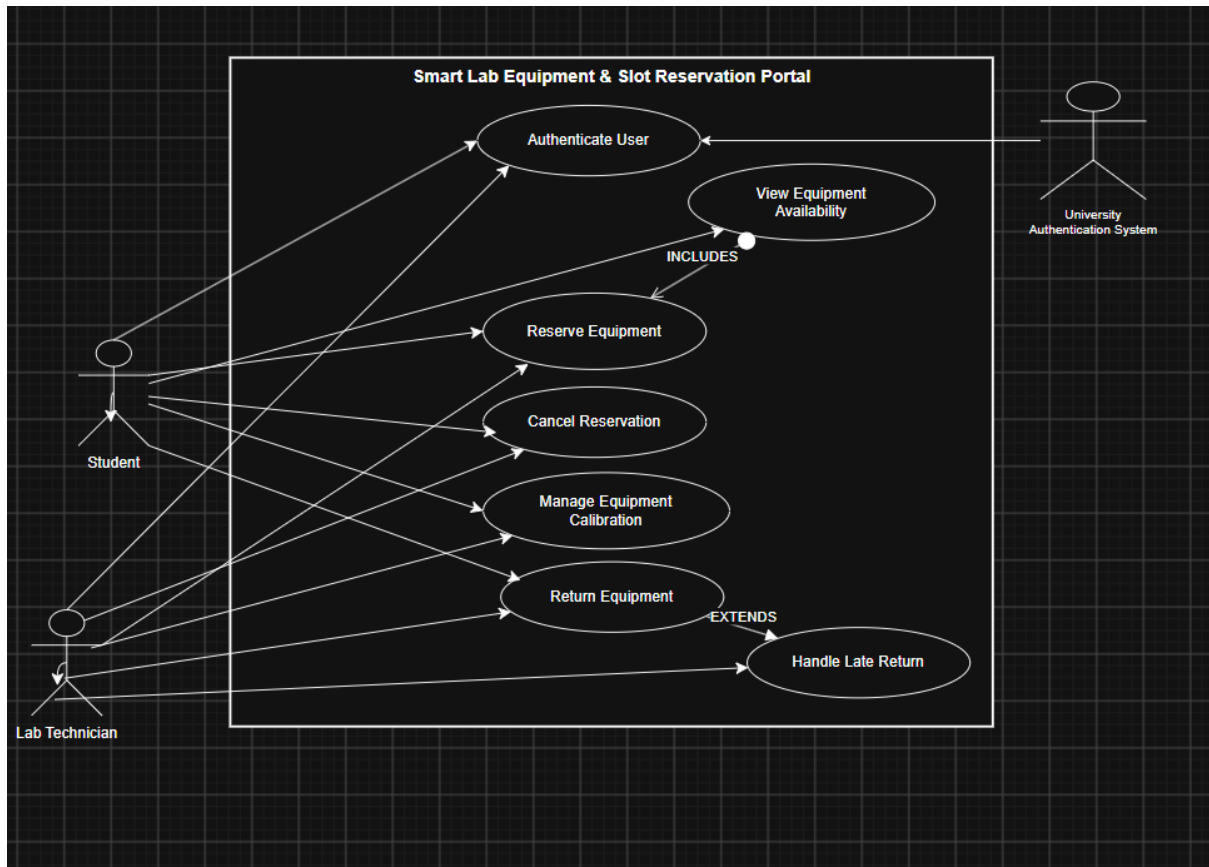


Req ID	type	description	priority	Acceptance Criteria	Rationale (short)
FR-001	functional	The system shall allow authenticated students to view real-time availability and reserve a maximum 2-hour slot for any calibrated equipment up to 7 days in advance.	High	Pass: An authenticated student can reserve a calibrated equipment slot of up to 2 hours within the next 7 days, and a reservation confirmation is generated. Fail: The system allows a reservation exceeding 2 hours, beyond 7 days, or for uncalibrated equipment.	Core functionality for controlled equipment reservation.
FR-002	functional	The system shall allow lab technicians to update the calibration status of laboratory equipment.	High	Pass: When a technician changes an equipment status to calibrated or uncalibrated, the updated status is immediately displayed in the equipment availability information. Fail: The system permits an uncalibrated equipment item to remain incorrectly marked as calibrated.	Ensures students can reserve only equipment with valid calibration status.
FR-003	functional	The system shall prevent students from reserving an equipment slot that is already	High	Pass: When a student attempts to reserve an occupied slot, the system rejects the	Prevents double-booking of laboratory equipment.

		reserved by another student.		reservation and displays that the slot is unavailable. Fail: Two students receive confirmed reservations for the same equipment and time slot.	
FR-004	functional	The system shall record equipment returns and identify reservations that exceed their scheduled return time.	High	Pass: When equipment is returned after its scheduled end time, the system records the late return and identifies the reservation as overdue. Fail: A late return is recorded as an on-time return.	Supports enforcement of late-return disciplinary rules.
FR-005	functional	The system shall allow students to cancel an active equipment reservation before its scheduled start time.	Medium	Pass: A student can cancel an eligible reservation and the corresponding equipment slot becomes available for reservation. Fail: A cancelled reservation remains active or the slot remains incorrectly unavailable.	Allows unused equipment slots to become available to other students.
NFR-001	NON-functional	The system shall ensure that concurrent slot reservation requests are processed within 200 ms	high	Pass: Performance testing confirms concurrent reservation requests are processed within 200 ms while	Prevents race conditions and double-booking during peak usage.

		with strict database-level lock isolation to prevent race conditions.		preventing conflicting reservations. Fail: Concurrent requests result in double-booking or exceed the 200 ms target.	
NFR-02	NON-functional	the system shall restrict equipment reservation and calibration-management operations to authenticated and authorized users.	high	Pass: An unauthenticated or unauthorized user attempting a restricted operation is denied access. Fail: An unauthorized user can create reservations or modify equipment calibration status.	Protects equipment and reservation information from unauthorized operations.

UML DIAGRAM



USE-CASE FLOW SPECIFICATION

Use Case ID: UC-03

Use Case Name: Reserve Equipment

Primary Actor: Student

Preconditions

1. Student is authenticated.
2. Equipment is registered in the system.
3. Equipment has a valid calibration status.
4. Student is eligible to make a reservation.

Postconditions

1. A reservation is created for the selected equipment and time slot.
2. The selected slot is locked against conflicting reservations.
3. A reservation confirmation/token is generated.
4. The reservation is associated with the student.

Main Success Scenario

1. Student selects **Reserve Equipment**.
2. System displays available laboratory equipment.
3. Student selects the required equipment.
4. System verifies the equipment's calibration status.
5. Student selects an available time slot.
6. System checks real-time slot availability.
7. System verifies that the requested reservation does not exceed 2 hours and is within the 7-day reservation window.
8. System locks the selected slot.
9. System creates the reservation.
10. System generates a reservation confirmation/token.

11. System displays the successful reservation confirmation to the student.

Alternate Flow: Selected Slot Becomes Unavailable

6a. The selected equipment slot is no longer available.

6a1. System informs the student that the selected slot cannot be reserved.

6a2. System does not create a reservation.

6a3. System displays other available slots.

6a4. Student may select another available slot.

This directly addresses the central problem of preventing double-booking.